

Migration, Crime and Education Analytics in Turkey

A Provincial Analysis (2008–2024)

Alp Çağrı Karapıçak Koray Ekici Habibe Akbaş
Mevlüt Faruk Dikmen Yahya Eren Kaya

May 2026

Table of contents

1	Abstract	2
2	Introduction	2
3	Data	2
4	Net Migration Analysis	5
4.1	Provincial Net Migration — 2024	5
4.2	Cumulative Net Migration — 2015–2024	7
5	Correlation and Regression Analysis	8
5.1	Hypotheses	8
5.2	Results	8
5.3	Regression Coefficients	8
5.4	Scatter Plots	9
6	Crime Trends in Major Provinces	10
7	Time Series Forecasting	10
7.1	Methodology	10
7.2	Forecast Plots	11
7.3	Point-Estimate Forecast Values	11
8	Crime Rate Heatmap	13
8.1	Methodology	13
8.2	Heatmaps	13
8.3	Key Findings	14
9	Conclusions	15

1 Abstract

This report analyses internal migration, crime, and education patterns across Turkey’s 81 provinces using administrative data from TÜİK (Turkish Statistical Institute) for the period 2008–2024. The study investigates: (1) the spatial distribution of net migration, (2) the statistical relationship between provincial crime counts and migration flows using Pearson correlation and linear regression, and (3) long-run crime trends and forecasts using ARIMA time series models. Key findings show a statistically significant positive correlation between in-migration and crime ($r = 0.359$, $p < 0.001$), a regression slope of approximately 3.4, and $R^2 = 0.13$. ARIMA forecasts indicate a continued upward trend in convictions across major provinces through 2030.

2 Introduction

Internal migration in Turkey is driven by economic, social, and environmental factors. Large metropolitan provinces such as İstanbul, Ankara, and İzmir attract substantial inflows from rural and smaller provinces, while simultaneously generating high crime statistics due to population density and urbanisation. Understanding whether and how migration correlates with crime is important for evidence-based regional policy.

The research questions addressed in this report are:

1. Which provinces experience the highest net migration gains and losses?
 2. Is there a statistically significant relationship between migration and crime at the provincial level?
 3. What are the projected crime trends to 2030 for selected provinces?
-

3 Data

All data are sourced from **TÜİK** (Türkiye İstatistik Kurumu). The datasets and their coverage are summarised below.

Dataset	File	Coverage
Provincial in-migration by reason	<code>aldigi_goc.xls</code>	81 provinces, 2018–2024
Provincial out-migration by reason	<code>verdigi_goc.xls</code>	81 provinces, 2018–2024
Educational attainment by province	<code>egitim.xlsx</code>	81 provinces, 2018–2024
Convicted persons by province	<code>suc.xls</code>	81 provinces, 2008–2020

Migration datasets contain 15 columns: year, province, total migration, and 12 reason-of-migration categories (appointment, finding work, education, family, housing, family-dependent, returning to hometown, health, house purchase, retirement, other, unknown).

4 Net Migration Analysis

4.1 Provincial Net Migration — 2024

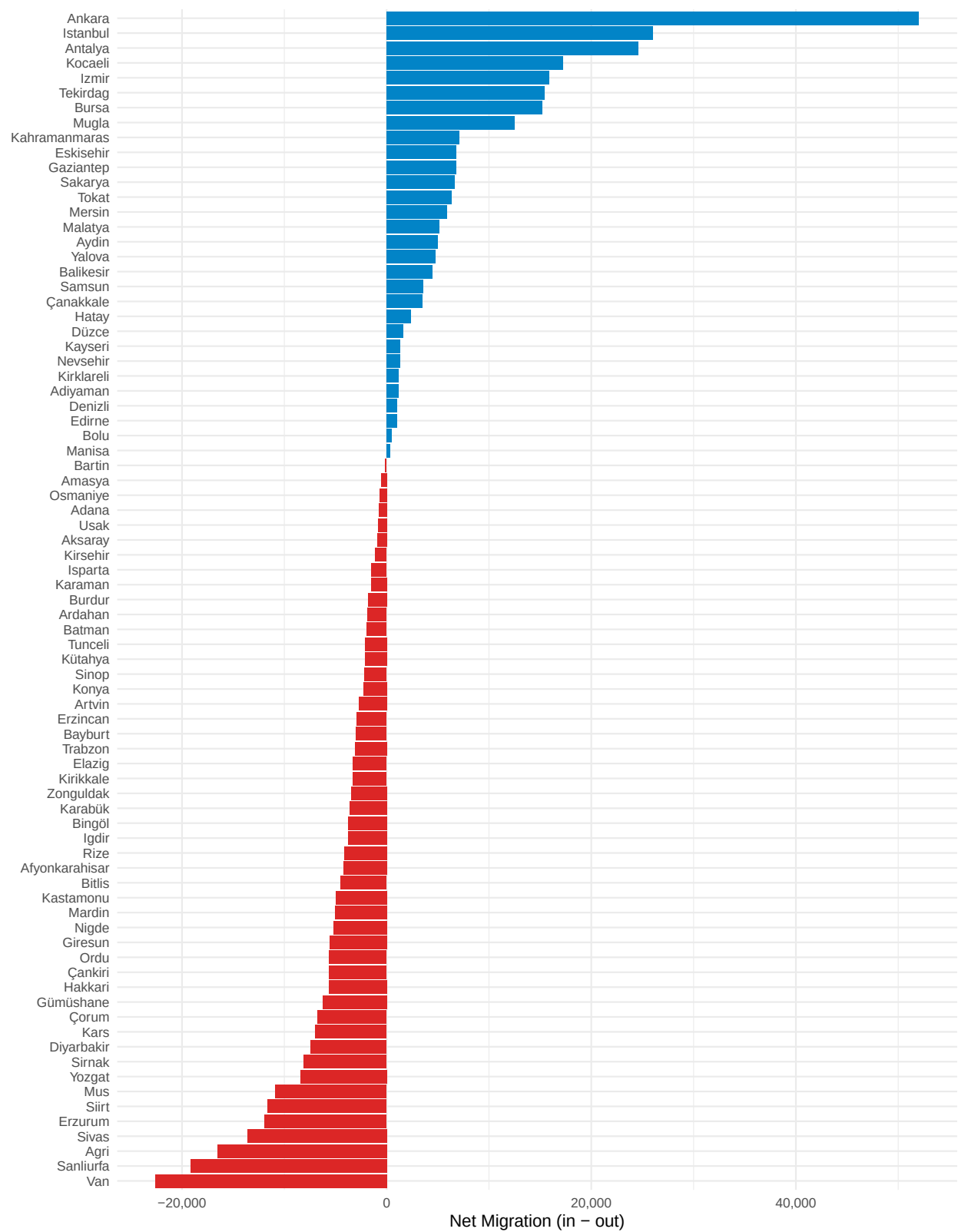


Figure 1: Net migration (in-migration minus out-migration) by province, 2024. Blue bars indicate net immigration; red bars indicate net emigration.

4.2 Cumulative Net Migration — 2015–2024

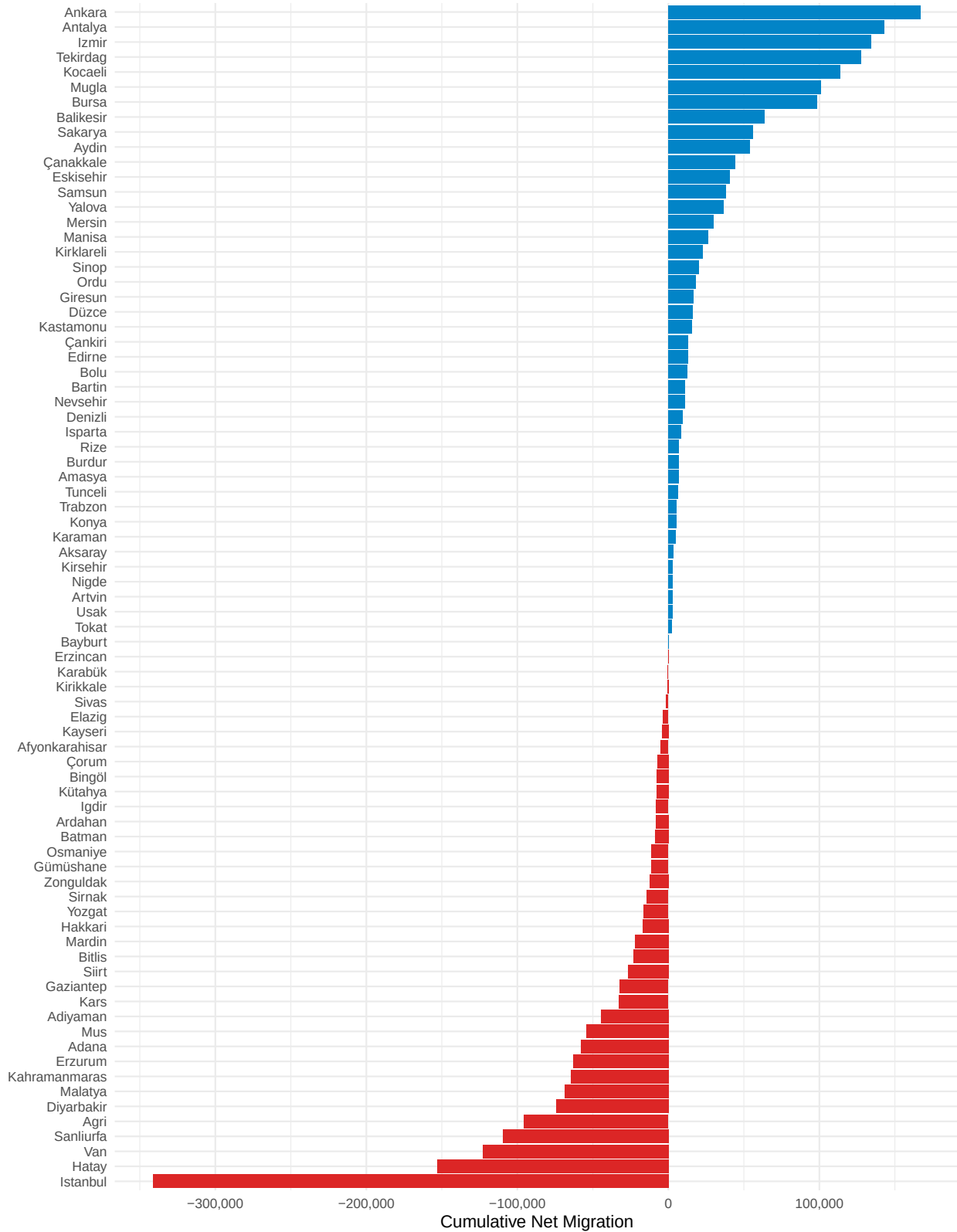


Figure 2: Cumulative net migration over the most recent decade. Provinces with large positive values are long-term migration attractors.

5 Correlation and Regression Analysis

5.1 Hypotheses

The central hypothesis for the migration–crime relationship is stated as follows:

	Hypothesis
H	There is <i>no</i> significant linear relationship between crime count and in-migration across provinces.
H	There <i>is</i> a significant linear relationship between crime count and in-migration across provinces.

The test is conducted at the 5% significance level ($\alpha = 0.05$).

5.2 Results

Table 1: Pearson Correlation: Migration vs. Crime Count

Comparison	Pearson r	t-statistic	p-value	95% CI Lower	95% CI Upper
In-Migration vs Crime	0.3525	5.700	3.66e-08	0.2341	0.4606
Out-Migration vs Crime	0.3544	5.734	3.07e-08	0.2361	0.4623

Since the p-value is much less than 0.05, **H is rejected**. A statistically significant positive relationship exists between crime count and in-migration ($r = 0.359$, $p < 0.001$).

The positive correlation indicates that provinces with higher crime counts tend to receive higher migration. However, the relationship is moderate: large, economically active provinces attract more migrants *and* record higher absolute crime counts due to population density and urbanisation effects.

5.3 Regression Coefficients

Table 2: Linear Regression: Migration ~ Crime Count

	Model	Term	Estimate	SE	t value	p-value	R ²
(Intercept)	In-Migration ~ Crime	(Intercept)	24074.000	3638.467	6.617	0	0.1243
SucSayisi		SucSayisi	3.117	0.547	5.700	0	NA

	Model	Term	Estimate	SE	t value	p-value	R ²
(Intercept)1	Out-Migration ~ Crime	(Intercept)	23081.732	3921.353	5.886	0	0.1256
SucSayisi1		SucSayisi	3.380	0.589	5.734	0	NA

The slope coefficient (~ 3.4) means that an increase of one additional conviction is associated with approximately 3.4 more in-migrants on average. The R^2 of ~ 0.13 indicates that crime count explains approximately **12.9%** of the variance in in-migration. This is statistically significant but indicates that crime is one of several socio-economic factors correlated with migration, not a dominant single cause.

5.4 Scatter Plots

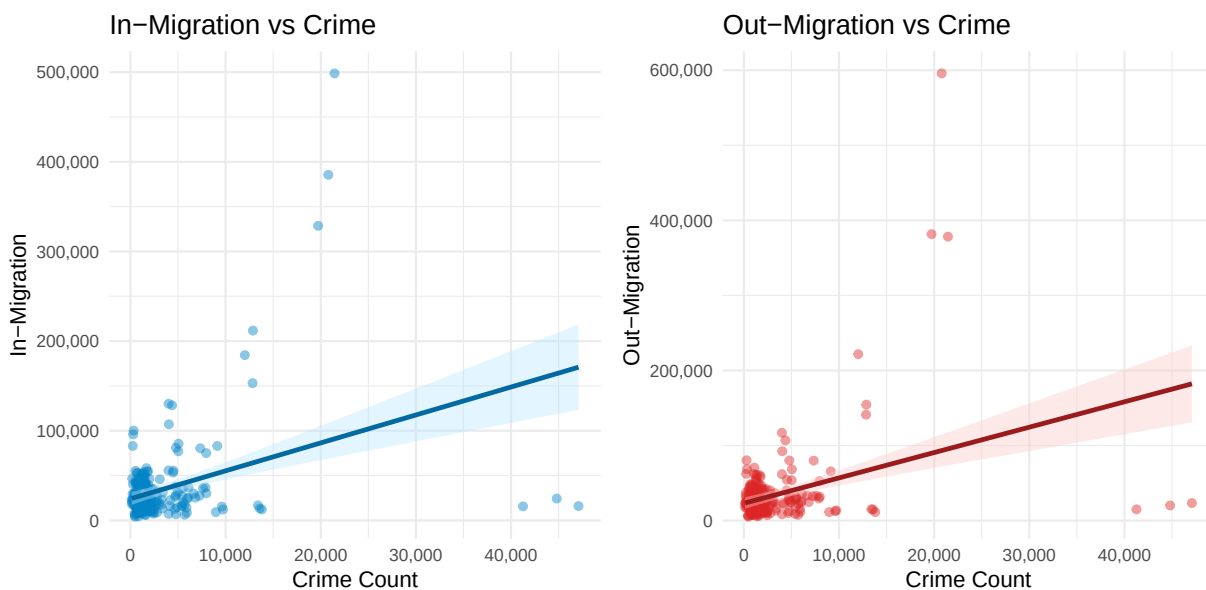


Figure 3: Scatter plots with OLS regression lines. Left: in-migration vs. crime. Right: out-migration vs. crime.

6 Crime Trends in Major Provinces

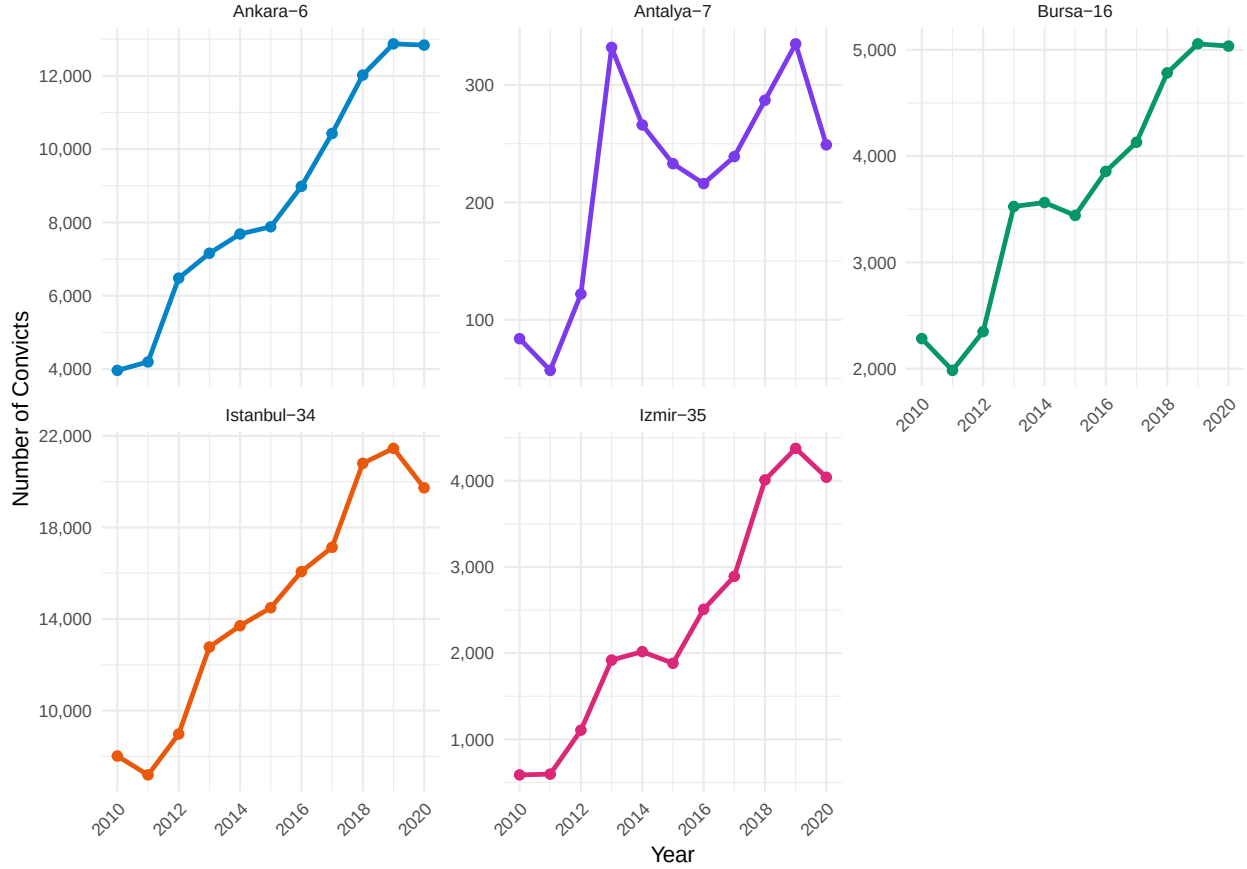


Figure 4: Annual number of convicted persons entering correctional institutions in five major provinces, 2010–2024.

7 Time Series Forecasting

7.1 Methodology

Forecasting was conducted using **ARIMA/SARIMA** models via the `auto.arima()` function from the `forecast` package in R. The function automatically selects the optimal model order based on the Akaike Information Criterion (AIC). The general SARIMA structure is:

$$\text{ARIMA}(p, d, q)(P, D, Q)_s$$

where p , d , q denote the non-seasonal autoregressive, differencing, and moving-average orders respectively; P , D , Q are their seasonal counterparts; and s is the seasonal frequency. Since the dataset consists of **annual observations**, seasonality is not applicable ($s = 1$), and models primarily capture long-run trend behaviour. Forecasts are presented with **80%** and **95%** confidence intervals.

7.2 Forecast Plots

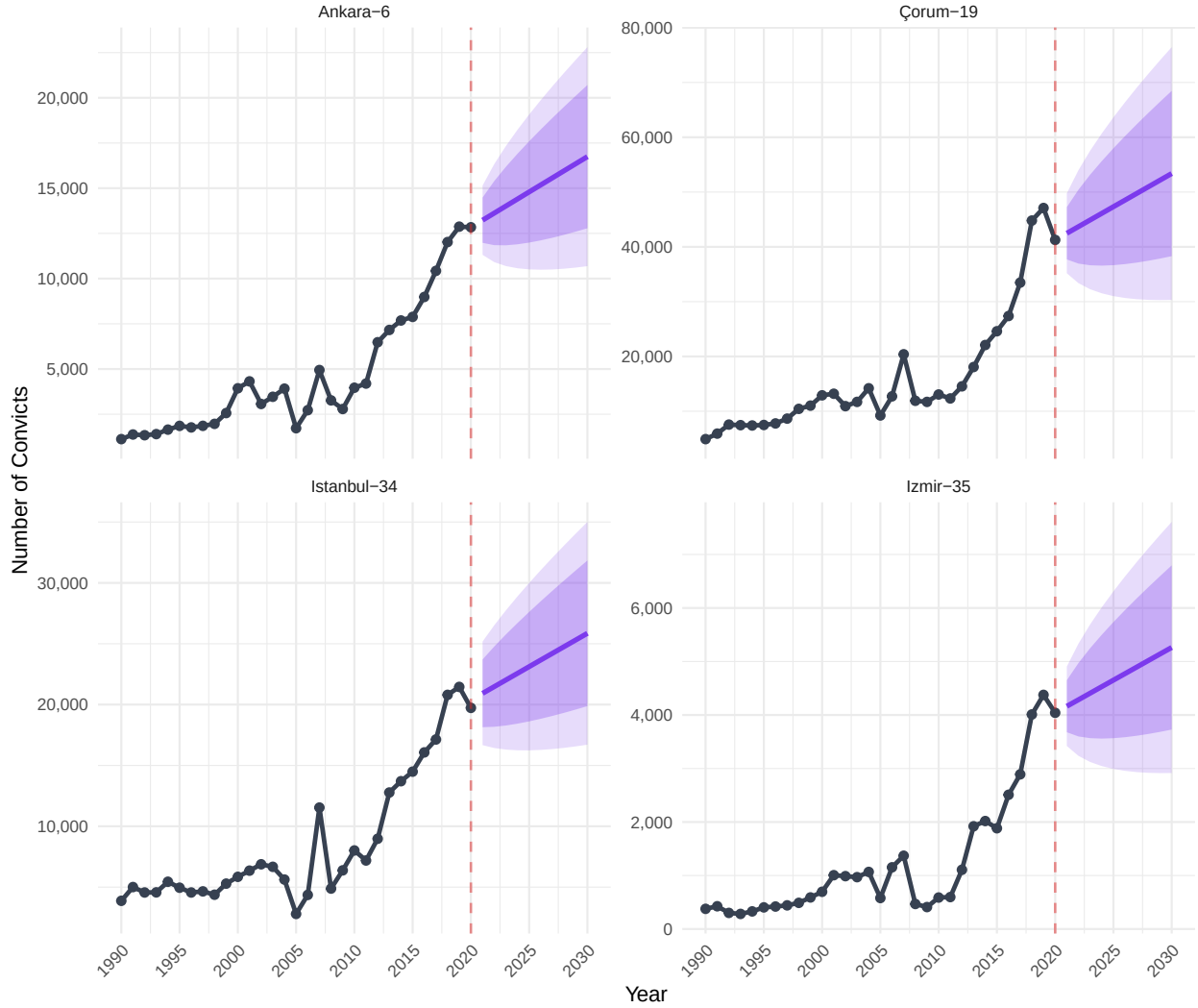


Figure 5: ARIMA crime forecasts to 2030 for four selected provinces. Shaded bands show 80% (dark) and 95% (light) confidence intervals. The dashed red line marks 2020.

7.3 Point-Estimate Forecast Values

Table 3: ARIMA Point-Estimate Forecasts — Convicted Persons

Province	2021	2025	2030
İstanbul	20,923	23,117	25,858
Ankara	13,231	14,793	16,746
İzmir	4,163	4,652	5,262
Çorum	42,478	47,325	53,385

Province	Trend	Predictability
İstanbul	Strong upward; accelerating after 2010	Moderate — wide CI
Ankara	Gradual, stable increase	High — narrow CI
İzmir	Moderate upward, flattening post-2020	Moderate
Çorum	Sharp increase after 2014; highest relative growth	Low — very wide CI

All provinces show an **increasing long-term trend**. The 2030 95% confidence interval for İstanbul is approximately 16,706–35,011, reflecting the substantial uncertainty inherent in long-range forecasting.

8 Crime Rate Heatmap

8.1 Methodology

Rather than raw conviction counts, this section uses **population-adjusted crime rates** (convictions per 100,000 population) to enable fair comparisons across provinces of different sizes:

$$\text{Crime Rate} = \frac{\text{Convictions}}{\text{Population}} \times 100,000$$

8.2 Heatmaps

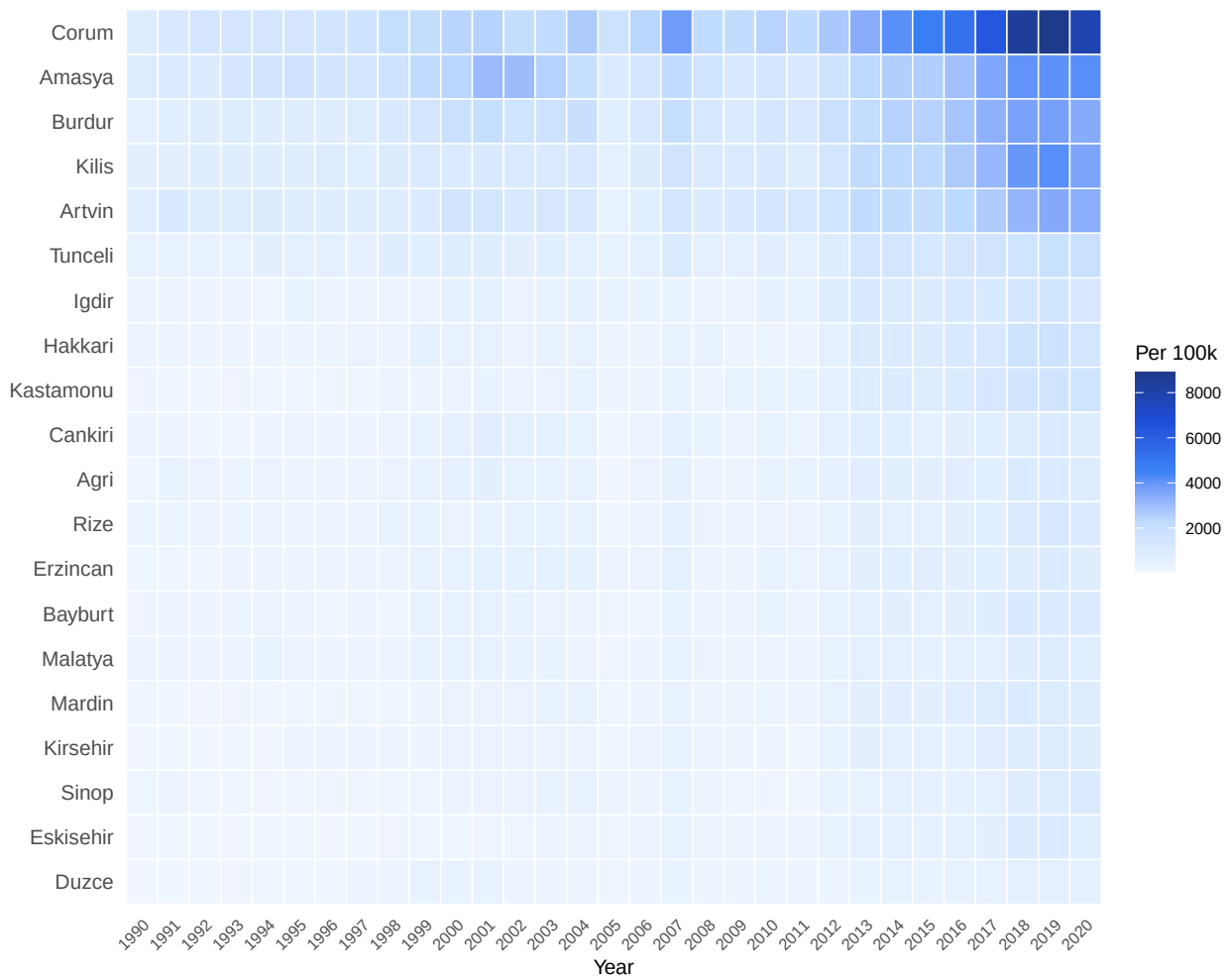


Figure 6: Crime rate heatmap (standard scale) — top 20 provinces by average conviction rate, 2008–2020.

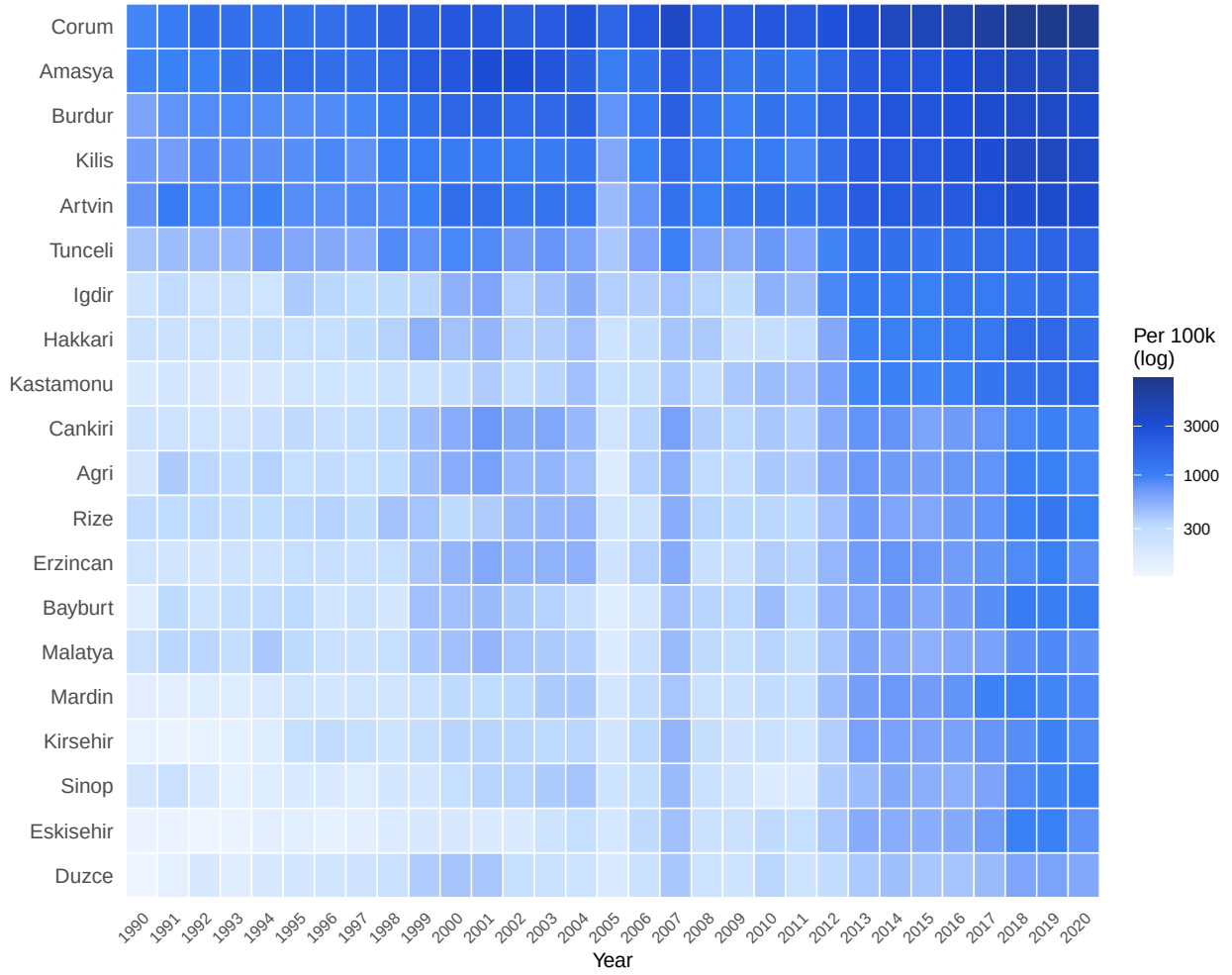


Figure 7: Crime rate heatmap (log scale) — compresses extreme values to reveal variation among mid-range provinces.

8.3 Key Findings

- Crime intensity is **not evenly distributed** across Turkey — provinces such as Çorum, Amasya, Burdur, Kilis, and Artvin show persistently high normalised crime rates.
- A **general upward trend** is visible after approximately 2013, with notably sharper increases between 2017 and 2020.
- **Smaller and medium-sized provinces can exhibit higher per-capita crime rates** than major metropolitan areas — demonstrating the importance of population normalisation.
- The log-scale transformation provides a more balanced view, making moderate regional differences clearly distinguishable.

9 Conclusions

1. **Net migration patterns** are highly polarised: İstanbul, Ankara, İzmir, and coastal tourist provinces consistently gain population, while many eastern and central Anatolian provinces experience net outflows.
2. **Migration and crime are positively correlated** ($r = 0.359$, $p < 0.001$). The regression slope (~ 3.4) and R^2 (~ 0.13) suggest a moderate but significant relationship, largely explained by shared underlying factors — urban size, economic activity, and population density — rather than a direct causal link.
3. **ARIMA forecasts** project a continued upward trend in convictions across all analysed provinces through 2030. Confidence intervals widen substantially over longer horizons, reflecting inherent uncertainty in trend extrapolation with annual data.
4. **Per-capita crime rates** reveal that smaller provinces are often disproportionately affected — a pattern obscured when raw counts are used — underscoring the need for population normalisation in comparative analyses.

Data source: TÜİK (Türkiye İstatistik Kurumu). Analysis period: 2008–2024. Hacettepe University — Department of Industrial Engineering — EMU 430 — Fall 2024.